

Report On Gold Loan Branch Performance Analysis and Segmentation using Data Analytics



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Abstract

Performance analysis of financial branches plays an important role in understanding operational efficiency, business growth, and risk exposure in lending institutions. In the gold loan industry, branches handle large volumes of loan transactions involving various parameters such as loan amount, pledged gold weight, loan-to-value (LTV) ratio, and rate of interest (ROI). Analyzing these parameters at the branch level can provide valuable insights into lending patterns, profitability, and risk management practices. This project focuses on analyzing gold loan disbursement data to evaluate branch performance and segment branches based on their operational performance and risk characteristics.

The study follows a structured analytical approach consisting of multiple stages. In the first stage, the loan-level dataset was cleaned and preprocessed to ensure data consistency and reliability. In the second stage, exploratory data analysis was performed to examine key loan characteristics such as loan disbursement distribution, scheme usage, state-wise loan activity, and variations in loan size. These analyses helped identify patterns in loan operations across different branches and regions. In the third stage, loan-level data was aggregated to generate branch-level performance metrics including total disbursement, loan count, average ROI, average LTV, and average gold weight. These metrics were used to evaluate the operational performance and lending behavior of each branch.

In the final stage, clustering techniques were applied to segment branches into different categories based on their business performance and risk levels. The clustering model grouped branches into four major segments: High Performance–Low Risk branches, High Performance–High Risk branches, Stable branches, and Underperforming branches. This segmentation provides a structured understanding of how branches differ in terms of loan volume, profitability, and lending risk.

To support data exploration and interpretation, an interactive analytics dashboard was developed to visualize key performance indicators, loan distributions, branch performance metrics, and cluster segmentation results. The dashboard enables users to analyze loan patterns, compare branch performance, and explore business insights through interactive visualizations.

The results demonstrate that branch performance analysis combined with data-driven segmentation can provide valuable insights for financial institutions. The analysis helps identify high-performing branches, detect high-risk lending practices, and highlight underperforming branches that require operational improvements. The findings highlight the potential of data analytics and clustering techniques in supporting data-driven decision-making and improving branch performance in gold loan operations.

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Chapter 1

Introduction

1.1 Introduction to the Project

Performance analysis is an important aspect of data-driven decision-making in financial institutions. In the gold loan industry, large volumes of loan transactions are generated across multiple branches, involving parameters such as loan amount, pledged gold weight, loan-to-value (LTV) ratio, and rate of interest (ROI). Analyzing these parameters helps organizations evaluate branch performance, understand lending patterns, and identify potential risk exposure. By studying historical loan data, financial institutions can gain insights into operational efficiency and improve their lending strategies.

Traditionally, branch performance has been evaluated using simple financial indicators such as loan disbursement and loan count. However, with the availability of large-scale financial data, advanced data analytics techniques can be used to perform more detailed analysis. By combining multiple performance metrics such as total disbursement, average ROI, average LTV, and loan count, a more comprehensive understanding of branch operations can be achieved.

In this project, gold loan disbursement data is analyzed to evaluate the performance of different branches. Loan-level transaction data is aggregated to generate branch-level performance metrics, which are then used to analyze operational behavior across branches. Clustering techniques are applied to segment branches into different categories based on performance and risk levels.

The branches are grouped into four segments: High Performance–Low Risk, High Performance–High Risk, Stable branches, and Underperforming branches. This segmentation provides a clearer understanding of branch performance and helps identify branches that generate strong business results as well as those that require improvement.

An interactive analytics dashboard is also developed to visualize key performance indicators, loan distributions, and branch segmentation results. The dashboard enables users to explore branch performance and gain insights through interactive visualizations.

Overall, this project demonstrates how data analytics and clustering techniques can be used to analyze branch performance, identify operational patterns, and support data-driven decision-making in gold loan operations.

1.2 Organization profile



This project is conducted in collaboration with **Indel Money Limited**, a leading Non-Banking Financial Company (NBFC) in India that offers various financial services including gold loans, MSME loans, consumer durable loans, and foreign exchange services. Established in 1986, Indel Money has grown into a trusted financial institution with a strong presence across India and a wide network of branches.

Gold loans are one of the major financial services provided by the company, allowing customers to pledge gold jewellery as collateral in exchange for short-term credit. Each branch handles multiple loan transactions involving key parameters such as loan amount, pledged gold weight, loan-to-value (LTV) ratio, and rate of interest (ROI).

With a large number of branches and loan transactions, analyzing operational data becomes important for evaluating branch performance and managing lending risk. Indel Money increasingly focuses on data-driven decision-making to improve operational efficiency and enhance its financial services.

In this context, analyzing loan disbursement data and branch-level performance metrics can provide valuable insights into business performance and risk exposure. Such analysis helps identify high-performing branches, detect potential risks, and support better strategic decision-making within the organization.

1.3 Objectives of the project

The primary objective of this project is to analyze gold loan disbursement data and evaluate the performance of different branches using data analytics techniques. The study focuses on identifying patterns in lending operations and segmenting branches based on their business performance and risk characteristics. The specific objectives of the project are as follows:

Objective1:

To study and analyze the gold loan disbursement dataset and understand key loan parameters such as loan amount, pledged gold weight, loan-to-value (LTV) ratio, and rate of interest (ROI).

Objective2:

To perform data preprocessing and exploratory data analysis in order to identify patterns in loan distribution across branches, schemes, and geographical regions.

Objective3:

To aggregate loan-level transaction data to generate branch-level performance metrics such as total disbursement, loan count, average ROI, average LTV, and average gold weight..

Objective4:

To apply clustering techniques to segment branches into different performance categories based on business volume, profitability, and risk exposure.

Objective5:

To develop an interactive analytics dashboard to visualize loan distributions, branch performance metrics, and clustering results for better understanding and decision-making.

Chapter 2

Dataset Description

2.1 Gold Loan Disbursement Dataset

2.1.1 Problem Statement

In the gold loan industry, a large number of loan transactions are processed across multiple branches, generating significant operational data. These transactions include key parameters such as loan amount, pledged gold weight, loan-to-value (LTV) ratio, and rate of interest (ROI). Analyzing this data is important for understanding lending patterns, evaluating branch performance, and identifying potential risk exposure.

The main problem addressed in this project is the difficulty in assessing branch performance and risk levels using raw loan transaction data. Without proper analysis, it becomes challenging for financial institutions to identify high-performing branches, detect high-risk lending behavior, and recognize underperforming branches.

This project aims to analyze the gold loan disbursement dataset and apply clustering techniques to segment branches based on performance and risk metrics. The analysis helps provide insights that can support better performance monitoring and data-driven decision-making in gold loan operations.

2.1.2 Dataset Definition

The dataset used in this project is the Gold Loan Disbursement Dataset, which contains loan transaction details collected from multiple branches of Indel Money. The dataset includes information related to individual gold loan transactions and the key parameters associated with each loan.

Number of rows: Large-scale transactional dataset containing thousands of loan records

Frequency: Transaction-based records

Row Description:

Each row in the dataset represents an individual gold loan transaction processed at a branch. The values include important information about the loan amount, pledged gold weight, loan-to-value ratio (LTV), rate of interest (ROI), and the branch where the loan was processed.

The dataset contains several columns including:

- Branch name
- Loan amount
- Net gold weight
- Gold price per gram
- Loan-to-value ratio (LTV)
- Rate of interest (ROI)
- Loan scheme
- State / branch location

This dataset is used to analyze loan disbursement patterns and to generate branch-level performance metrics for further analysis and segmentation.

2.1.3 Collection of Data

The dataset used in this project was obtained from Indel Money's central data repository, which contains operational records of gold loan transactions across its branch network.

The data was provided in Excel format and used for analysis and branch performance evaluation. More information about the organization is available at <https://indelmoney.com/>.

2.1.4 Strategy of Collecting the Data

The gold loan disbursement dataset was selected because it represents real-world financial transaction data collected across multiple branches of Indel Money. The dataset contains important lending parameters such as loan amount, pledged gold weight, loan-to-value (LTV) ratio, and rate of interest (ROI).

Using this dataset enables the analysis of branch performance, lending patterns, and risk exposure, and supports the segmentation of branches based on their operational performance.

2.2 Branch Performance Dataset

2.2.1 Problem Statement

In a financial institution with multiple branches, evaluating the operational performance of each branch is essential for effective business management and risk control. Branches handle numerous loan transactions, making it difficult to directly compare performance using raw transactional data alone. Without structured analysis, identifying branches that generate strong business performance or those that carry higher lending risk becomes challenging. This dataset is used to assess branch performance and to provide structured metrics that can support performance comparison, risk evaluation, and further clustering-based segmentation of branches.

2.2.2 Dataset Definition

The Branch Performance Dataset is generated by aggregating the gold loan transaction data at the branch level. It summarizes key operational metrics that represent the performance of each branch within the lending network.

Number of rows: Approximately equal to the number of branches in the dataset

Frequency: Branch-level aggregated records

Row Description:

Each row in the dataset represents the aggregated performance metrics of a specific branch. The values summarize loan transactions handled by the branch and provide insights into its business volume, lending behavior, and risk exposure.

The dataset contains several columns including:

- Branch name
- Total disbursement amount
- Loan count
- Average rate of interest (ROI)
- Average loan-to-value ratio (LTV)
- Average gold weight pledged

This dataset is used to evaluate branch-level operational performance and serves as the primary dataset for branch comparison and clustering-based segmentation.

2.2.3 Collection of Data

The branch performance dataset was created using the gold loan disbursement dataset obtained from Indel Money's central data repository. Loan-level transaction data was aggregated to generate branch-level performance metrics.

The processed dataset was prepared and stored in Excel format for analysis and visualization. More information about the organization can be accessed at <https://indelmoney.com/>.

2.2.4 Strategy of Collecting the Data

The branch performance dataset was developed by aggregating loan transaction data to create meaningful branch-level metrics. This approach allows the evaluation of business performance, profitability, and risk exposure across different branches.

Using this dataset enables structured analysis of branch operations and supports further segmentation of branches based on their performance characteristics

Chapter 3

Data Preprocessing and Exploratory Data Analysis (Gold Loan Disbursement Dataset)

Data preprocessing and exploratory data analysis (EDA) are essential steps in any data analytics project. These processes help in cleaning the dataset, handling missing or inconsistent values, and preparing the data for further analysis.

In this project, preprocessing was performed on the gold loan disbursement dataset to ensure data quality and consistency. Exploratory data analysis was then conducted to understand key patterns in loan disbursement, branch activity, and lending parameters such as loan amount, LTV ratio, and ROI. This analysis provides important insights into the structure of the data and supports further branch performance evaluation and clustering-based segmentation.

3.1 Data Cleaning and Preprocessing

Data preprocessing is an important step in data analytics that involves cleaning and transforming raw data into a structured format suitable for analysis. Since the gold loan dataset contains transaction-level records from multiple branches, preprocessing was necessary to ensure data consistency and accuracy before performing branch-level analysis. The preprocessing steps included data cleaning, handling missing values, converting non-numeric data types, and preparing the dataset for aggregation.

3.1.1 Data Cleaning and Data Type Conversion

The initial dataset contained several columns with inconsistent formats and non-numeric values that required conversion before analysis. Certain categorical variables such as ESGIN status were converted into numeric format to enable easier processing and aggregation. Additionally, columns representing numerical values such as loan amount, net gold weight, gold price per gram, LTV ratio, and ROI were verified and converted to appropriate numeric data types. This ensured that calculations and statistical analysis could be performed correctly.

3.1.2 Handling Missing Values

The dataset was examined for missing or incomplete entries that could affect the analysis. Missing values in important fields were identified and addressed to maintain data integrity. In cases where missing values were minimal, appropriate cleaning techniques were applied to remove or correct the entries. This step ensured that the dataset remained consistent and reliable for further analysis.

3.1.3 Branch-Level Data Aggregation

After cleaning and preprocessing the dataset, the loan-level transaction data was aggregated to generate branch-wise performance metrics. Key indicators such as total disbursement amount, loan count, average ROI, average LTV, and average gold weight were calculated for each branch. This aggregation transformed the raw transactional data into a structured branch-level dataset that could be used for performance evaluation and clustering-based segmentation.

3.2 Exploratory Data Analysis

Exploratory Data Analysis (EDA) is performed to understand the structure of the dataset and identify patterns, trends, and relationships among different variables. Through visualizations and summary statistics, EDA helps reveal important insights about loan disbursement behavior across states, schemes, and branches.

In this project, several visualizations were created to analyze loan disbursement patterns, scheme usage, and geographical distribution of loans. These visualizations provide a clearer understanding of how gold loan operations vary across different regions and financial schemes. Prophet uses a decomposable time series model based on a generalized additive model framework. This formulation allows the model to capture nonlinear trends and multiple seasonalities such as weekly or yearly cycles. The model also supports changepoints that allow the trend to adjust when structural changes occur in the data.

3.2.1 Top States by Disbursement

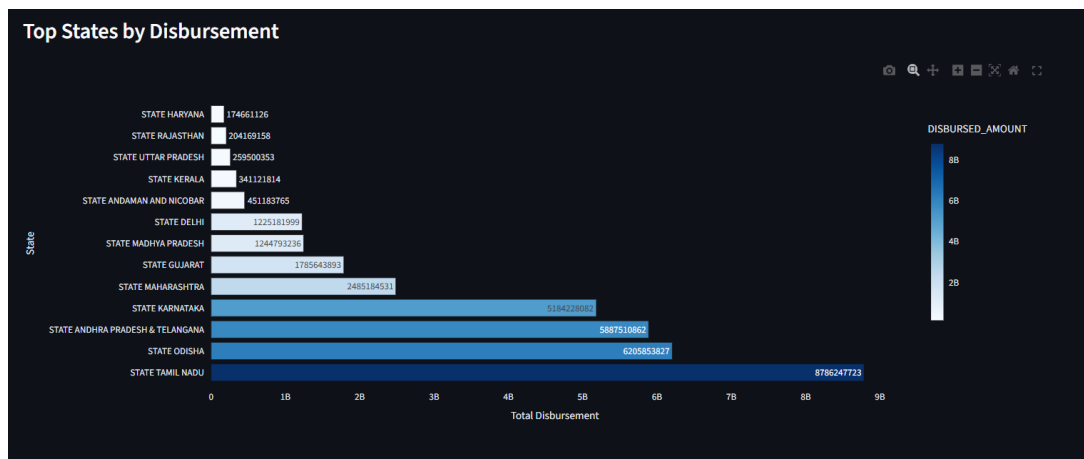


Figure 3.2.1: Total loan disbursement across different states highlighting the top contributing regions.

This visualization presents the total loan disbursement amount across different states. The chart shows that Tamil Nadu contributes the highest total disbursement, followed by Odisha and Andhra Pradesh & Telangana. States such as Karnataka and Maharashtra also contribute significantly to the overall loan disbursement. On the other hand, states like Haryana and Rajasthan have relatively lower disbursement amounts. This indicates that the demand for gold loans is more concentrated in certain regions, particularly in southern states where gold loans are widely used for short-term financial needs.

3.2.2 Disbursement by Scheme

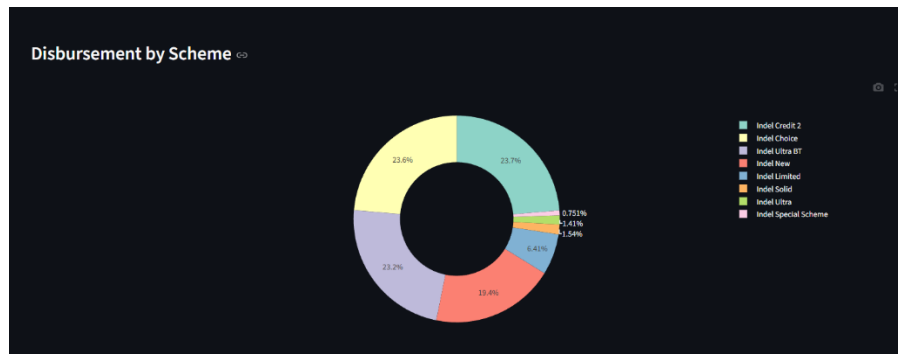


Figure 3.2.2: Distribution of total loan disbursement across different gold loan schemes.

The donut chart illustrates the proportion of total loan disbursement across different gold loan schemes offered by the organization. The visualization shows that Indel Credit 2, Indel Choice, and Indel Ultra BT contribute the largest share of total disbursement. Schemes such as Indel New and Indel Limited also account for a considerable portion of the loan volume. Meanwhile, schemes like Indel Solid, Indel Ultra, and Indel Special Scheme represent a smaller portion of the total disbursement. This distribution highlights which loan schemes are most preferred by customers and which schemes generate the highest loan volume.

3.2.3 Number of Loans by Scheme

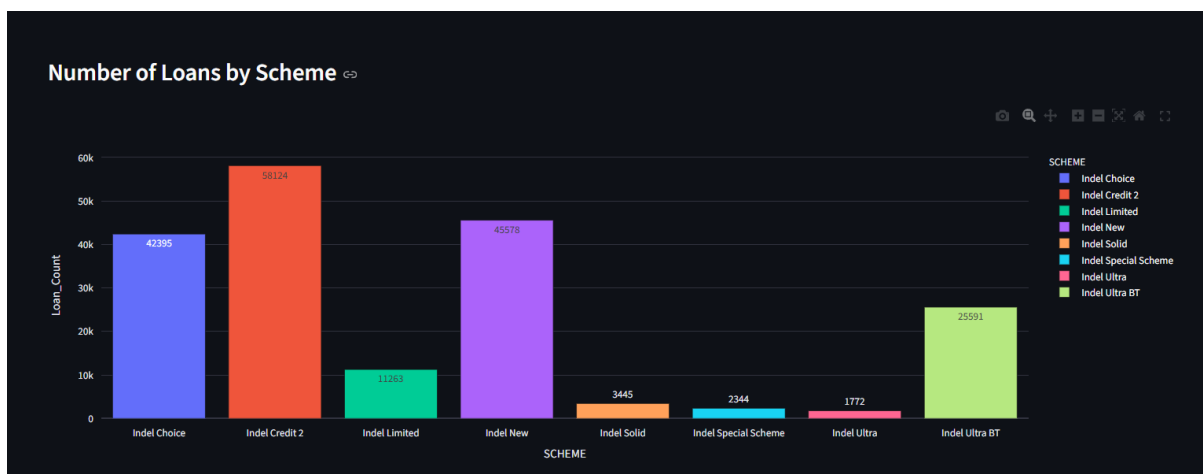


Figure 3.2.3: Number of loan transactions issued under each gold loan scheme.

This visualization shows the total number of loans issued under each scheme. The chart indicates that Indel Credit 2 has the highest number of loan transactions, followed by Indel New and Indel Choice. Schemes such as Indel Ultra BT also account for a significant number of loans. In contrast, Indel Solid, Indel Special Scheme, and Indel Ultra have comparatively fewer transactions. This analysis helps identify which schemes are most frequently used by customers and which ones are less commonly utilized.

3.2.4 State Contribution to Total Disbursement

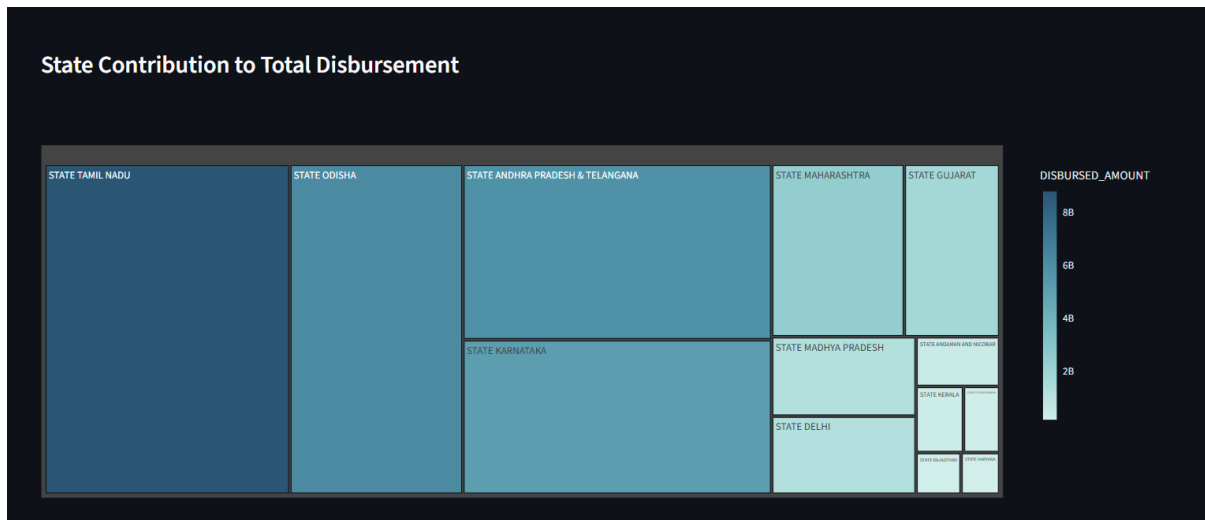


Figure 3.2.4: Treemap visualization showing the contribution of each state to the total loan disbursement.

The treemap visualization illustrates the contribution of each state to the total loan disbursement. Larger blocks represent states that contribute a higher share of the total disbursement amount. From the chart, Tamil Nadu occupies the largest portion, indicating its dominant contribution to the company's loan portfolio. Other states such as Odisha, Andhra Pradesh & Telangana, Karnataka, and Maharashtra also show significant contributions. Smaller blocks correspond to states with relatively lower disbursement amounts. This visualization helps understand the geographical distribution of gold loan activity across different regions.

3.2.5 Loan Distribution by State and Scheme

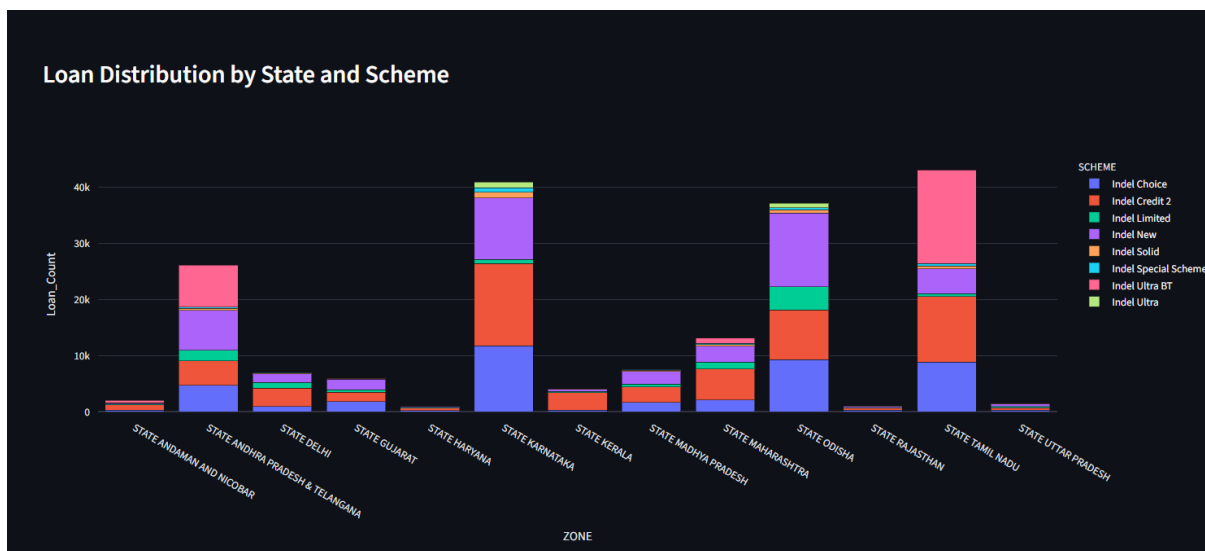


Figure 3.2.5: Distribution of loan transactions across states categorized by different loan schemes.

This stacked bar chart shows the distribution of loan transactions across different states and schemes. The visualization highlights how various loan schemes are utilized across different regions. States such as Tamil Nadu, Karnataka, and Odisha have a higher number of loans across multiple schemes, indicating strong loan activity. The chart also reveals

that certain schemes dominate in specific states, suggesting regional preferences for particular loan products. This analysis helps identify how loan schemes are distributed geographically and which schemes perform well in different states.

3.2.6 Average Loan Size by Scheme

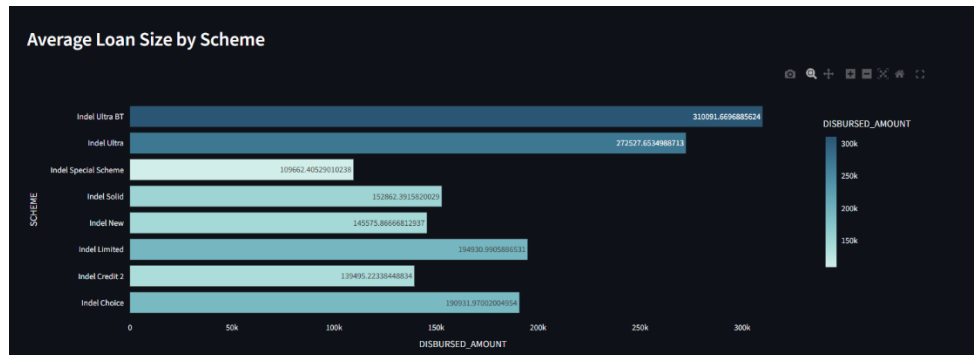


Figure 3.2.6: Average loan size issued under each gold loan scheme.

This chart presents the average loan size issued under each scheme. The visualization shows that Indel Ultra BT has the highest average loan size, followed by Indel Ultra and Indel Limited. Schemes such as Indel Choice and Indel Credit 2 have moderate average loan sizes, while Indel Special Scheme has a relatively lower average loan amount. This analysis helps understand the lending characteristics of different schemes and highlights which schemes are associated with larger loan disbursements.

3.3 Correlation and Feature Relationship Analysis

Correlation and feature relationship analysis helps in understanding how different variables in the dataset interact with each other and influence loan disbursement patterns. In this study, relationships between key numerical variables such as loan amount, net gold weight, gold price per gram, loan-to-value (LTV) ratio, and rate of interest (ROI) were examined to understand how these factors contribute to loan disbursement decisions. For example, there is a natural relationship between net gold weight and loan amount, as higher pledged gold weight generally results in higher loan eligibility. Similarly, the LTV ratio determines the proportion of loan provided relative to the value of pledged gold.

During the exploratory analysis, it was also observed that loan schemes vary significantly across different regions and states. The percentage of total disbursement under each scheme changes depending on the geographical location and branch operations. Because of this regional variation, loan scheme cannot be considered a dominant feature for overall analysis, as its impact is not consistent across all regions. Instead, evaluating performance based on branch-level metrics provides a more accurate representation of operational performance.

Therefore, rather than relying heavily on scheme-level analysis, the study focuses on branch-wise aggregated metrics such as total disbursement, loan count, average ROI, and average LTV. This approach provides a more reliable understanding of lending behavior and allows more meaningful comparison of branch performance across the organization.

Chapter 4

Data Preprocessing and Exploratory Data Analysis (Branch Wise Performance Dataset)

Branch-level analysis provides a clearer understanding of how different branches contribute to the overall business performance of the organization. While loan-level transaction data provides detailed information about individual loans, aggregating this data at the branch level enables better evaluation of operational efficiency, business volume, and risk exposure.

In this stage of the project, the cleaned loan transaction dataset was aggregated to generate branch-wise metrics such as total disbursement, loan count, average ROI, average LTV, and average pledged gold weight. These metrics allow meaningful comparison between branches and help identify high-performing branches, branches with high lending activity, and branches with higher risk exposure. The following analysis explores these branch-level metrics through various visualizations.

4.1 Data Cleaning and Preprocessing

Before performing branch-level analysis, the aggregated dataset was further examined to ensure that the calculated metrics were accurate and consistent. The preprocessing stage included validating aggregated values, checking for outliers in branch metrics, and confirming that numerical columns such as total disbursement, loan count, average ROI, average LTV, and average gold weight were correctly computed.

4.1.1 Verification of Aggregated Metrics

After converting the transaction-level dataset into branch-level metrics, the aggregated values were verified. This included confirming that total disbursement values matched the sum of loan amounts processed by each branch and that the loan count correctly represented the number of transactions handled by the branch.

Key checks performed included:

- Verification of total disbursement calculations
- Validation of loan count values
- Cross-checking of average ROI and LTV calculations
- Ensuring numerical consistency across all aggregated columns

This step ensured that the aggregated dataset accurately represented the operational activity of each branch.

4.1.2 Handling Inconsistent Values

During preprocessing, the dataset was examined for extreme values or inconsistencies in the calculated branch metrics. Variables such as average ROI, average LTV, and average gold weight were carefully reviewed to ensure that no abnormal values were present due to data aggregation errors.

This step ensured:

- Removal or correction of inconsistent entries

- Validation of realistic value ranges for ROI and LTV
- Accurate representation of branch performance metrics

4.1.3 Preparation for Visualization and Analysis

After verifying and cleaning the aggregated dataset, it was prepared for visualization and exploratory analysis. The final branch-level dataset contained structured metrics that allowed the creation of analytical visualizations to explore performance patterns across the branch network.

The prepared dataset included the following important features:

- Branch name
- Total disbursement amount
- Loan count
- Average ROI
- Average LTV
- Average gold weight pledged

These metrics formed the foundation for further branch performance analysis and clustering-based segmentation.

variables.

4.1 Exploratory Data Analysis

Exploratory Data Analysis (EDA) on the branch-level dataset helps reveal patterns in branch performance, loan activity, and lending behavior. By analyzing aggregated metrics such as disbursement volume, loan count, ROI, and LTV, it becomes possible to identify differences in branch operations and understand how various factors influence lending performance.

The following visualizations provide insights into branch-level business performance, operational efficiency, and relationships between key lending variables.

4.2.1 Top Branches by Disbursement

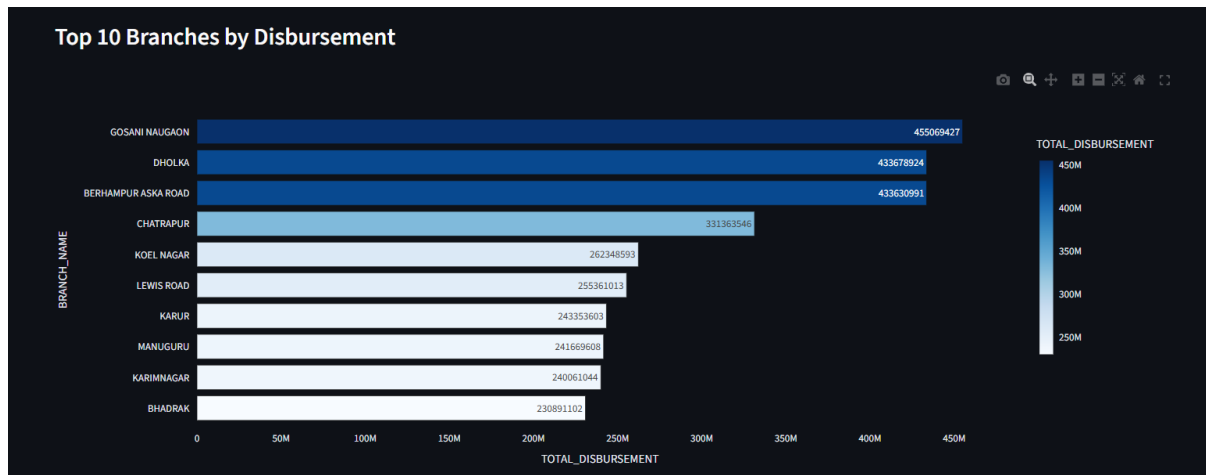


Figure 4.2.1: Top branches ranked by total loan disbursement amount.

This visualization highlights the branches with the highest total loan disbursement. These branches represent locations where lending activity and business volume are particularly strong.

Key observations:

- Gosani Nuagaon records the highest disbursement among all branches.
- Dholka and Berhampur Aska Road also contribute significantly to the total loan portfolio.
- High disbursement branches indicate areas with strong demand for gold loans.
- These branches play a critical role in driving overall revenue and lending activity.

This analysis helps identify key revenue-generating branches within the organization.

4.2.2 Branch Risk vs Performance



Figure 4.2.2: Relationship between average LTV and total disbursement across branches.

This scatter plot illustrates the relationship between average LTV and total disbursement across branches, with colour representing average ROI. Branches with higher

disbursement volumes generally operate within a moderate LTV range. The visualization helps identify how lending risk and business performance interact across different branches.

Important insights include:

- Branches with higher disbursement volumes typically operate within moderate LTV ranges.
- Extremely high LTV values could indicate higher lending risk exposure.
- Color variations represent average ROI, showing how profitability differs between branches.
- The visualization highlights how branches balance risk (LTV) and business performance (disbursement).

This chart helps identify branches that may combine high performance with potentially higher risk levels.

4.2.3 Top Branches by Loan Count

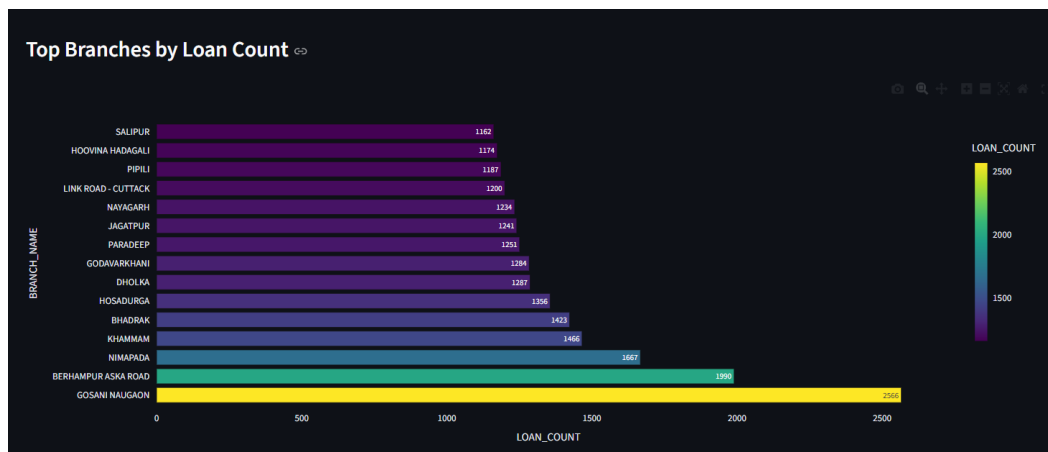


Figure 4.2.3: Branches with the highest number of loan transactions.

This chart presents the branches that handle the highest number of loan transactions. Branches such as Gosani Nuagaon and Berhampur Aska Road appear among the most active branches in terms of loan processing. High loan counts indicate strong customer engagement and operational activity.

Key insights:

- Gosani Nuagaon appears as the branch with the highest loan count.
- Berhampur Aska Road and Nimapada also handle a large number of transactions.
- High loan count indicates strong customer demand and active branch operations.
- Branches with high loan counts often serve large customer bases.

This analysis helps understand branch workload and operational intensity.

4.2.4 ROI Distribution Across Branches

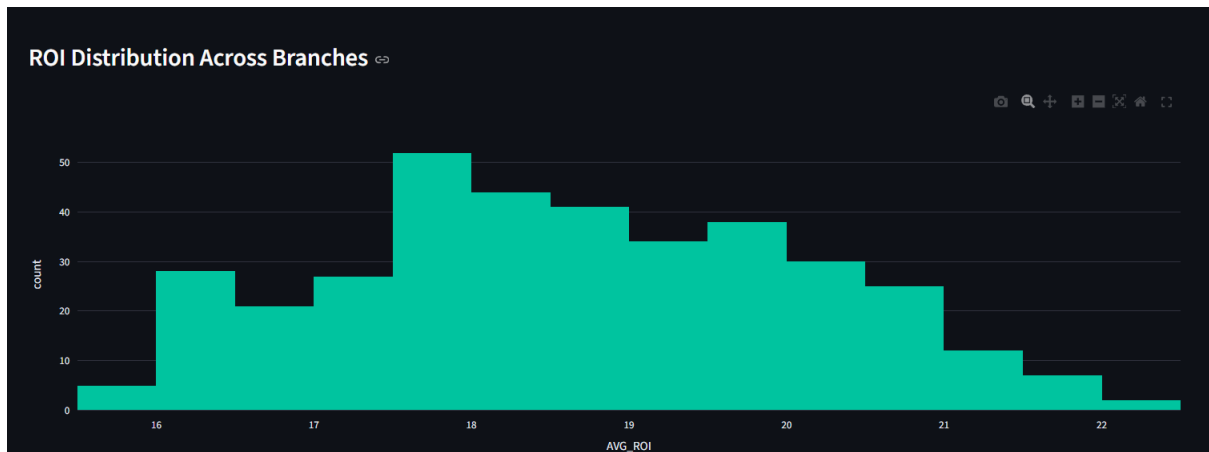


Figure 4.2.4: Distribution of average ROI values across branches.

The histogram shows how average interest rates vary across different branches.

Important observations:

- Most branches operate within a ROI range of approximately 17% to 21%.
- This indicates relatively consistent interest rate policies across branches.
- Some branches show slightly higher ROI values, suggesting differences in lending strategies or scheme selection.

This visualization helps understand profitability patterns across the branch network.

4.2.5 LTV Distribution Across Branches

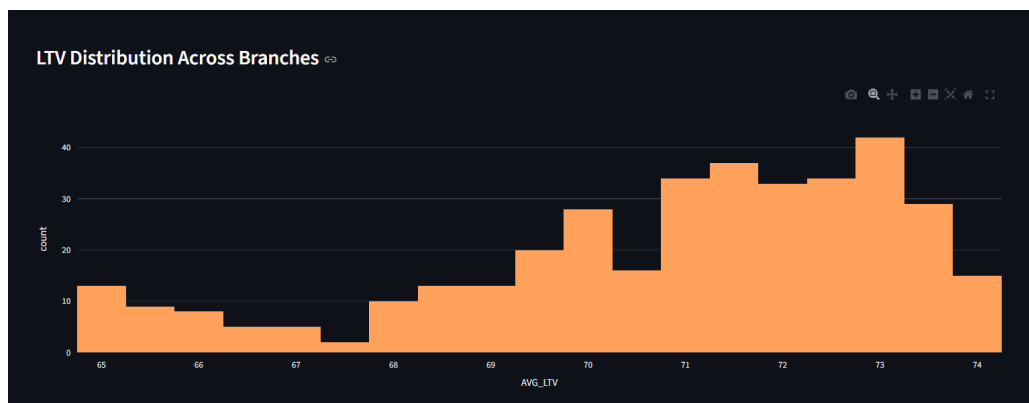


Figure 4.2.5: Distribution of average LTV values across branches.

This visualization presents the distribution of average LTV ratios across branches. Most branches operate within a relatively narrow LTV range, suggesting consistent lending practices across the organization. Variations in LTV may indicate differences in risk tolerance or branch-level loan approval practices.

Key insights:

- Most branches operate within an LTV range between 69% and 73%.

- This suggests standardized lending policies across the organization.
- Slight variations may reflect local market conditions or branch-level decision making.

This analysis helps evaluate risk distribution across the branch network.

4.2.6 Gold Weight vs Disbursement

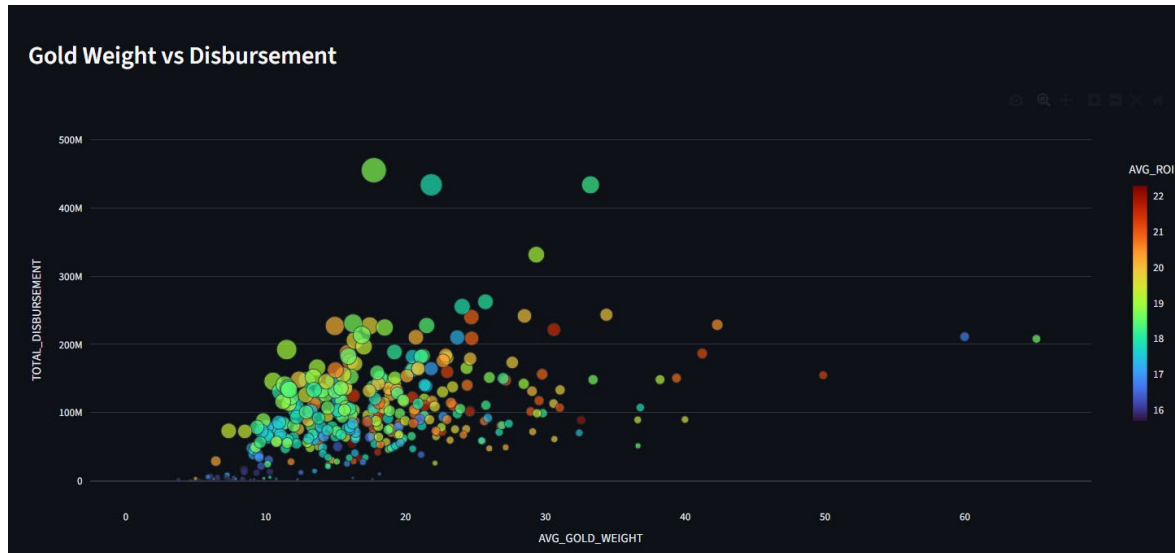


Figure 4.2.6: Relationship between pledged gold weight and total disbursement.

This scatter plot shows the relationship between average gold weight pledged and total disbursement. A positive relationship can be observed, indicating that higher pledged gold weight generally leads to larger loan disbursements. The color gradient represents average ROI, highlighting how interest rates vary across branches. Key insights include:

- A positive relationship can be observed between gold weight and loan disbursement.
- Branches with higher pledged gold weight typically generate larger loan amounts.
- Color variation indicates average ROI, highlighting how interest rates vary across branches.

This confirms that gold weight is a key determinant of loan disbursement.

4.2.7 Loan Count vs ROI

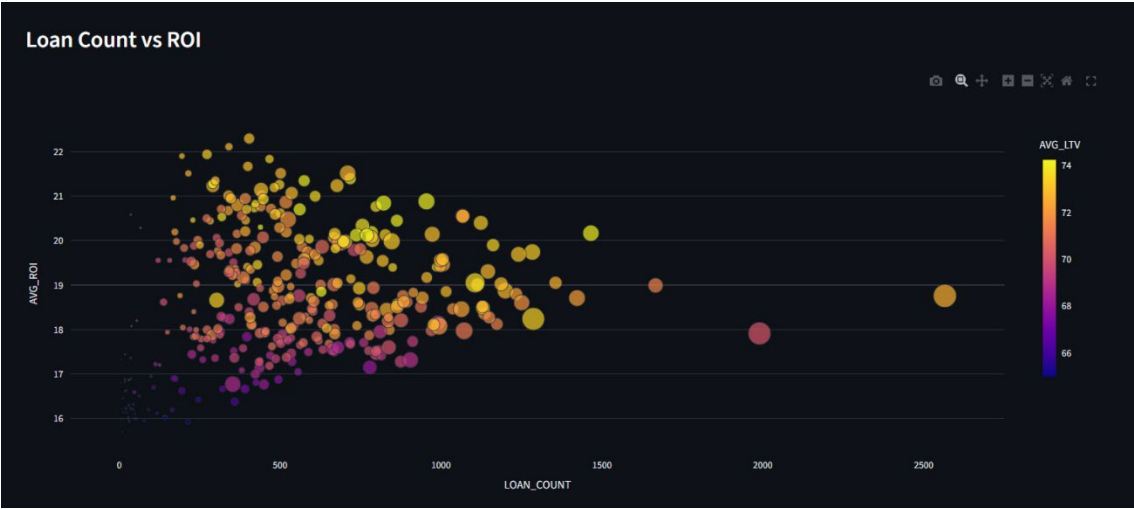


Figure 4.2.7: Relationship between loan count and average ROI across branches.

This visualization examines the relationship between loan count and average ROI. Branches with higher loan counts tend to operate within moderate ROI ranges, indicating balanced lending strategies. The color gradient represents average LTV, providing insight into how lending risk varies with loan activity. Important observations:

- Branches with higher loan counts often operate with moderate ROI levels.
- Extremely high ROI values are not always associated with high loan counts.
- Colour variation represents average LTV, indicating how risk levels vary across different branches.

This chart helps understand how loan volume, profitability, and risk interact across branches.

4.2.8 Top Branches by ROI

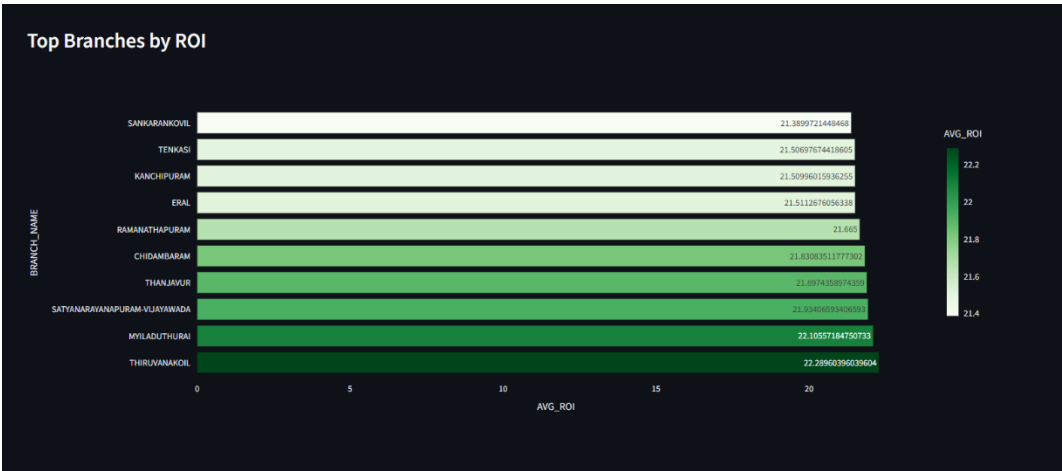


Figure 4.2.8: Branches ranked by highest average ROI.

This chart highlights the branches that achieve the highest average ROI values. Branches such as Thiruvananthapuram, Mayiladuthurai, and Satyanarayanapuram-Vijayawada appear among the top performers in terms of profitability. Higher ROI values indicate stronger revenue generation for the organization..

Key insights:

- Thiruvananthapuram and Mayiladuthurai appear among the branches with the highest ROI values.
- Higher ROI may indicate specific loan schemes or branch-level pricing strategies.
- These branches contribute significantly to profit generation within the organization.

This analysis helps identify branches that generate higher revenue through interest earnings.

4.3 Correlation and Feature Relationship Analysis

Correlation and feature relationship analysis was conducted to understand how different variables in the dataset interact with each other and influence loan disbursement patterns. Key numerical variables such as loan amount, net gold weight, loan-to-value ratio (LTV), rate of interest (ROI), and total disbursement were examined to identify relationships that affect lending behavior. Understanding these relationships helps in selecting the most relevant variables for further analysis and performance evaluation.

Based on the exploratory and correlation analysis, three primary factors were identified as the most important indicators for evaluating branch performance and conducting clustering analysis. These factors capture the major dimensions of branch operations and provide a balanced view of performance and risk.

The three key factors selected for clustering analysis are:

1. Profitability – Average Rate of Interest (ROI)

The average ROI represents the profitability generated by a branch through its lending operations. Branches with higher ROI values contribute more to interest-based revenue, making ROI an important indicator of financial performance.

2. Risk Exposure – Loan-to-Value Ratio (LTV)

The LTV ratio reflects the level of lending risk associated with loan disbursement. Higher LTV values indicate that a larger portion of the gold value is being lent to customers, which may increase the risk exposure of the branch.

3. Business Volume – Total Disbursement

Total disbursement represents the overall lending activity and business volume handled by a branch. Branches with higher disbursement amounts indicate stronger loan demand and higher operational activity.

These three variables profitability (ROI), risk exposure (LTV), and total disbursement were identified as the most significant features for evaluating branch performance. Therefore, they were selected as the primary inputs for the clustering model used in the subsequent stage of the project to segment branches based on performance and risk characteristics.

Chapter 5

Branch Performance Clustering Model

Machine learning techniques provide powerful tools for analyzing large datasets and identifying hidden patterns within data. In financial institutions, clustering techniques are commonly used to group branches based on similar operational characteristics. Such segmentation allows organizations to better understand branch performance, identify high-performing branches, and detect branches that may require operational improvements.

In this project, a clustering-based machine learning approach was applied to segment branches using branch-level performance metrics derived in the previous chapter. The clustering process involved several stages including feature selection, feature scaling, dimensionality reduction using Principal Component Analysis (PCA), determination of the optimal number of clusters, and implementation of the K-Means clustering algorithm. The resulting clusters represent groups of branches with similar operational patterns in terms of business volume, profitability, and lending risk.

5.1 Feature Selection for Clustering

Feature selection is an important step in machine learning because it determines which variables will be used to perform clustering analysis. Based on the exploratory data analysis and correlation analysis conducted in previous chapters, three major factors were identified as the most important indicators of branch performance.

The selected features represent three major dimensions of branch operations:

Profitability – Average Rate of Interest (ROI)

The average rate of interest (ROI) represents the profitability generated by a branch through its lending activities. Branches with higher ROI values contribute more to interest-based revenue and therefore indicate stronger financial performance.

Risk Exposure – Loan-to-Value Ratio (LTV)

The loan-to-value ratio (LTV) represents the level of lending risk associated with loans issued by a branch. Higher LTV values indicate that a larger portion of the pledged gold value is lent to customers, which increases potential financial risk.

Business Volume – Total Disbursement

Total disbursement represents the overall lending activity handled by a branch. Branches with higher disbursement amounts indicate stronger business performance and higher operational activity.

These three variables — ROI, LTV, and Total Disbursement — provide a balanced representation of profitability, risk exposure, and operational scale, making them suitable for clustering analysis.

5.2 Feature Scaling using StandardScaler

The selected variables have different numerical ranges. For example, total disbursement values may be in hundreds of millions, while LTV and ROI values are expressed as percentages. Without normalization, variables with larger numerical values could dominate the clustering process.

To address this issue, StandardScaler was applied to standardize the dataset. StandardScaler transforms each feature so that it has:

- Mean equal to 0
- Standard deviation equal to 1

This ensures that all features contribute equally to the clustering algorithm and improves

5.3 Dimensionality Reduction using Principal Component Analysis (PCA)

After scaling the data, Principal Component Analysis (PCA) was applied to reduce the dimensionality of the dataset and remove redundancy among correlated variables.

PCA transforms the original variables into a new set of variables called principal components, which capture the maximum variance present in the dataset.

The explained variance obtained from PCA is shown below:

Principal Component Variance Explained

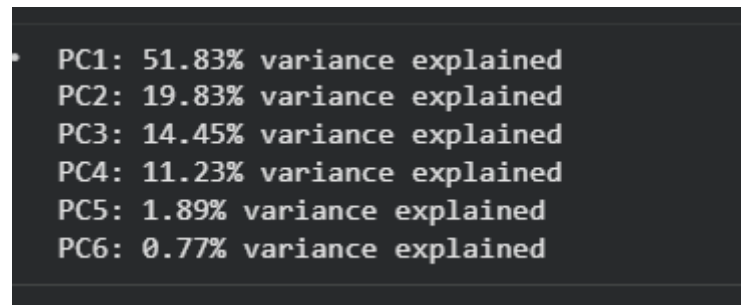


Figure 5.3.1: Explained variance of principal components obtained from PCA.

From the results, the first three principal components capture approximately 86% of the total variance in the dataset, indicating that most of the important information can be represented using these components.

The use of PCA helps:

- Reduce dimensionality
- Remove feature correlations
- Improve clustering performance
- Simplify visualization of clusters

5.4 Determining the Optimal Number of Clusters

Before applying the clustering algorithm, it is necessary to determine the optimal number of clusters.

Two evaluation techniques were used:

- **Elbow Method**
- **Silhouette Score Analysis**

5.4.1 Elbow Method

The Elbow Method determines the optimal number of clusters by analyzing the inertia values for different cluster counts.

Inertia represents the sum of squared distances between data points and their assigned cluster centroids. As the number of clusters increases, inertia decreases.

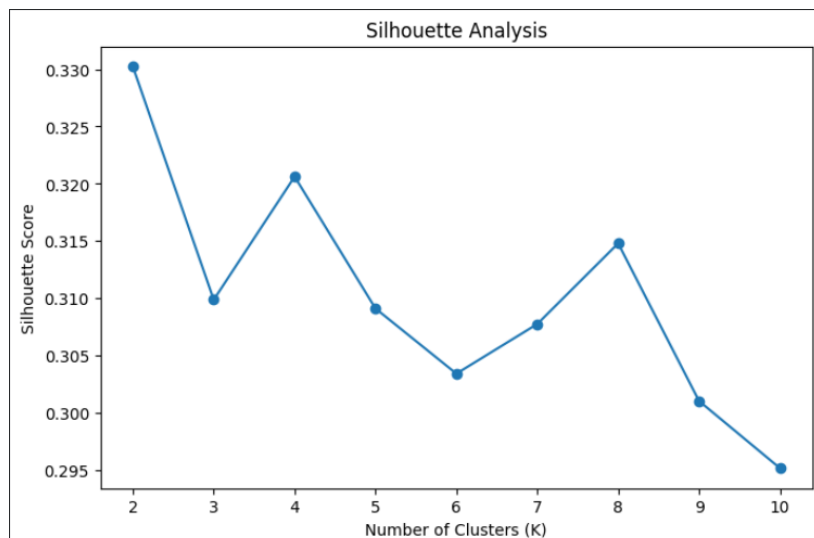


Figure 5.4.1: Elbow method used to determine the optimal number of clusters.

From the elbow plot, a noticeable bend occurs around **K = 4**, indicating that four clusters provide a good balance between model simplicity and clustering accuracy.

5.4.2 Silhouette Score Analysis

To validate the cluster selection, Silhouette Score analysis was also performed.

The silhouette score measures how similar a data point is to its own cluster compared to other clusters. The value ranges from -1 to 1, where:

- Values close to 1 indicate well-separated clusters
- Values near 0 indicate overlapping clusters
- Negative values indicate incorrect clustering

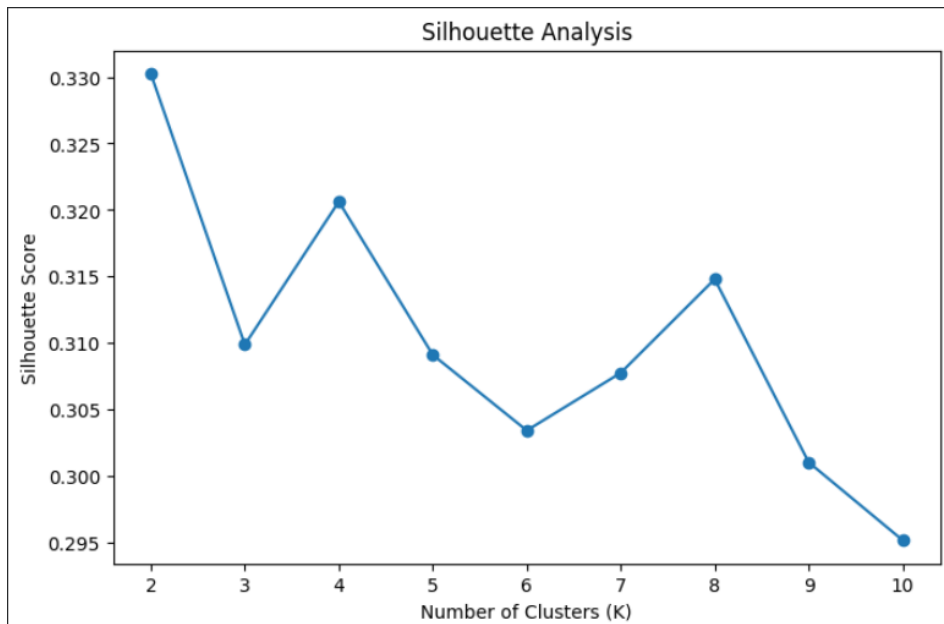


Figure 5.4.2: Silhouette score analysis for evaluating clustering quality.

The silhouette score results also support the use of four clusters, confirming the suitability of this cluster configuration.

5.5 K-Means Clustering Algorithm

After determining the optimal number of clusters, the K-Means clustering algorithm was applied to the dataset.

K-Means is an unsupervised learning algorithm that groups data points into clusters based on similarity. The algorithm minimizes the distance between data points and their respective cluster centroids.

The algorithm follows these steps:

1. Select the number of clusters (K).
2. Initialize cluster centroids randomly.
3. Assign each data point to the nearest centroid.
4. Recalculate cluster centroids.
5. Repeat the process until cluster assignments stabilize.

In this project, the K-Means algorithm was applied using **K = 4 clusters**, resulting in the segmentation of branches into four distinct groups.

5.6 Cluster Visualization

To interpret the clustering results, several scatter plots were created to visualize the relationships between the key variables.

Risk vs Volume by Cluster

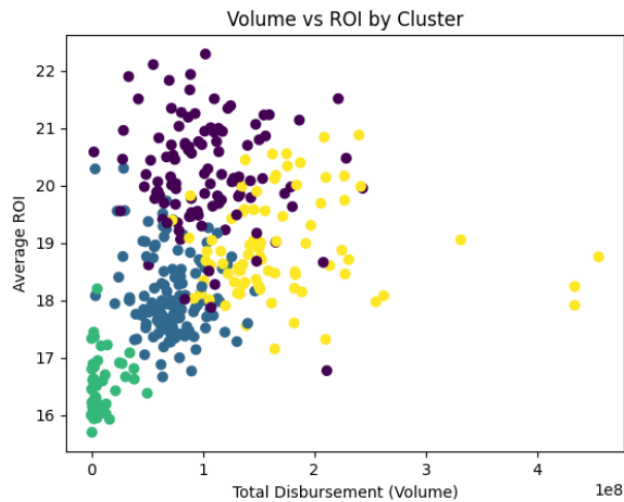


Figure 5.6.1: Relationship between lending risk (Average LTV) and business volume (Total Disbursement) across clusters.

This visualization shows how branches are distributed based on lending risk and disbursement volume.

Key observations include:

- Branches with higher disbursement volumes generally operate with higher LTV levels.
- Some clusters represent high-volume branches, while others represent low-volume branches.
- The visualization clearly separates branches based on their operational performance.

Risk vs ROI by Cluster

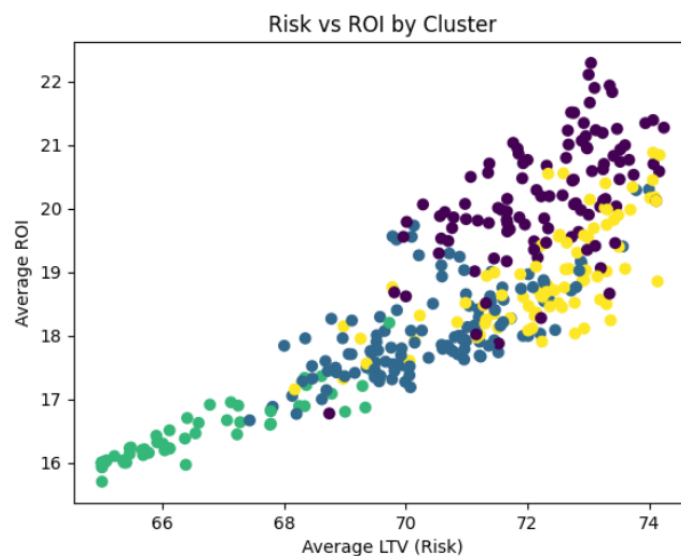


Figure 5.6.2: Relationship between lending risk (Average LTV) and profitability (Average ROI) across clusters.

This scatter plot shows the relationship between risk exposure and profitability.

Important insights include:

- A positive relationship can be observed between LTV and ROI.
- Branches operating with higher risk levels tend to generate higher profitability.
- Different clusters represent branches with different combinations of risk and profitability.

Volume vs ROI by Cluster

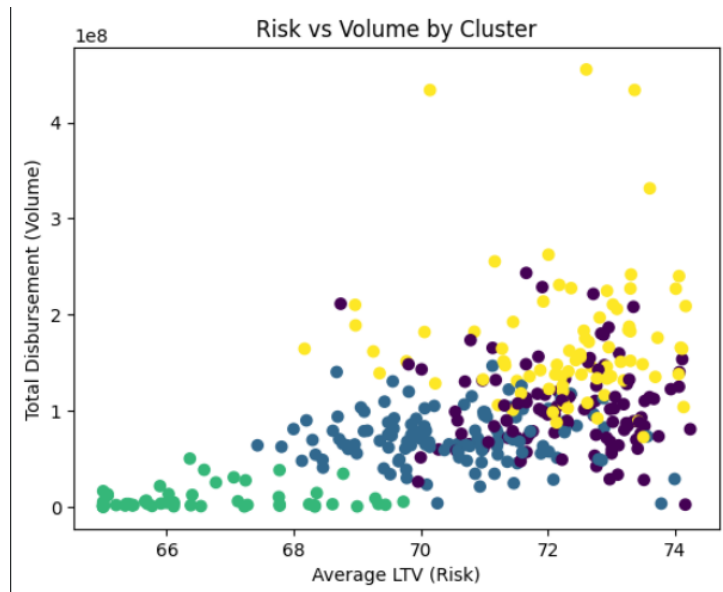


Figure 5.6.3: Relationship between business volume (Total Disbursement) and profitability (Average ROI) across clusters

This visualization illustrates how loan disbursement volume relates to profitability.

Key observations include:

- High-volume branches generally show higher ROI values.
- Some clusters represent high-performing branches with strong volume and profitability.
- Other clusters represent branches with lower activity levels.

5.7 Interpretation of Clusters

After clustering the branches, each cluster was analyzed based on the average values of ROI, LTV, and Total Disbursement.

The clusters can be interpreted as follows:

Cluster 1 – High Performance, High Risk

Branches in this cluster show high loan disbursement volumes and relatively high LTV values, indicating strong lending activity but higher risk exposure.

Cluster 2 – High Performance, Balanced Risk

Branches in this group demonstrate high business volume with moderate LTV levels and strong profitability, representing efficient branch operations.

Cluster 3 – Stable Branches

Branches in this cluster show moderate disbursement levels with balanced risk and profitability, indicating consistent operational performance.

Cluster 4 – Underperforming Branches

Branches in this group have lower disbursement volumes and lower ROI values, suggesting reduced lending activity and potential scope for improvement.

Chapter 6

Dashboard Development and visualization

Data visualization plays an important role in transforming raw analytical results into meaningful insights that can be easily understood by decision-makers. While statistical analysis and machine learning models provide valuable outputs, visual dashboards help present these insights in an interactive and intuitive manner. For financial institutions such as gold loan companies, dashboards enable management to monitor operational performance, identify trends, and make informed strategic decisions.

In this project, an interactive analytics dashboard was developed using Python Streamlit, a framework designed for building web-based data applications. The dashboard integrates the results of data preprocessing, exploratory data analysis, branch-level performance evaluation, and machine learning-based clustering. The platform allows users to explore the gold loan dataset through multiple analytical views and visualizations.

The dashboard is organized into different sections including Project Overview, Loan Level Analytics, Branch Performance Dashboard, and Cluster Analysis Dashboard. These sections provide a comprehensive view of the operational performance of gold loan branches across different regions.

6.1 Dashboard Overview

The main landing page of the application provides a summary of the project and the key analytical components included in the platform.

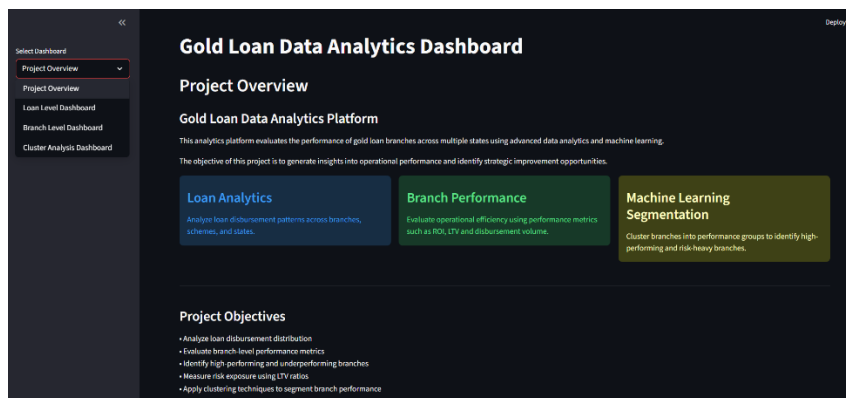


Figure 6.1.1: Main dashboard interface showing the project overview page.

The overview section introduces the Gold Loan Data Analytics Dashboard, which serves as a centralized platform for analyzing loan disbursement data across multiple branches and states. The dashboard highlights three major analytical components:

- **Loan Analytics** – analysis of loan disbursement patterns across branches, schemes, and states
- **Branch Performance Analysis** – evaluation of operational efficiency using performance indicators such as ROI, LTV, and disbursement volume
- **Machine Learning Segmentation** – clustering branches into performance groups using machine learning techniques

The overview page also lists the main objectives of the project, including analyzing loan distribution, evaluating branch performance, identifying high-performing branches, measuring risk exposure, and applying clustering techniques for branch segmentation.

The navigation panel on the left side of the dashboard allows users to switch between different analytical modules.

6.2 Loan Level Analytics Dashboard

The Loan Level Analytics Dashboard provides insights into individual loan transactions and overall lending patterns within the organization.

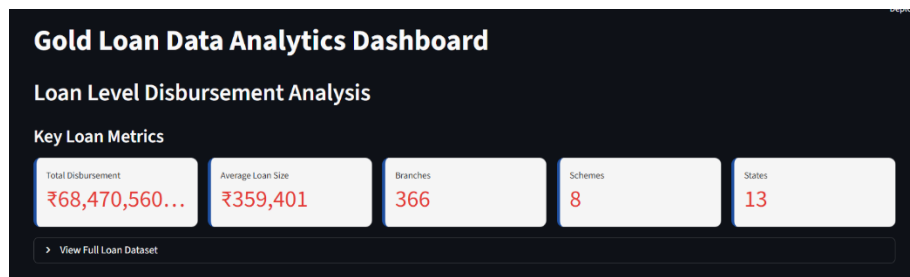


Figure 6.2.1: Loan level disbursement analytics dashboard displaying key loan metrics.

This dashboard includes several key indicators that summarize the loan dataset, including:

- **Total Disbursement Amount** – the total value of loans issued across all branches
- **Average Loan Size** – the average loan amount issued per transaction
- **Total Number of Branches** – the number of branches included in the dataset
- **Total Loan Schemes** – the number of different loan schemes offered
- **Total States Covered** – the number of states where loan operations are conducted

These summary metrics provide a quick overview of the scale and distribution of gold loan operations.

The loan-level dashboard allows users to explore loan data across different schemes, states, and branches. It helps identify lending patterns and understand how loan products are utilized across different regions.

Additionally, the dashboard includes an option to view the full loan dataset, enabling users to examine detailed transaction-level information.

6.3 Branch Performance Dashboard

The Branch Performance Dashboard focuses on evaluating the operational efficiency of individual branches. This dashboard uses aggregated branch-level metrics that were derived during the data preprocessing stage.

The key metrics analyzed in this section include:

- Total Disbursement Volume
- Average Rate of Interest (ROI)
- Average Loan-to-Value Ratio (LTV)

- Loan Count
- Average Gold Weight per Loan

These metrics help measure branch performance from three important perspectives:

1. **Business Volume** – measured through total loan disbursement
2. **Profitability** – represented by average ROI
3. **Risk Exposure** – represented by average LTV ratio

Through visual charts and comparisons, the branch dashboard helps identify:

- High-performing branches
- Branches generating high disbursement volumes
- Branches operating with higher risk levels
- Branches with lower operational performance

This section provides valuable insights for management to monitor branch efficiency and identify areas for improvement.

6.4 Cluster Analysis Dashboard

The Cluster Analysis Dashboard visualizes the results of the machine learning-based branch segmentation model developed in Chapter 5.

The clustering model grouped branches into different categories based on three primary performance indicators:

- Average ROI (profitability)
- Average LTV (risk exposure)
- Total Disbursement (business volume)

The cluster analysis dashboard displays scatter plots and visualizations that show how branches are distributed across these variables.

These visualizations allow users to easily identify clusters representing:

- High-performing branches
- Branches with high lending risk
- Stable branches with moderate activity
- Underperforming branches

By visualizing the clustering results, the dashboard helps decision-makers quickly understand how branches differ in terms of performance and operational behavior.

This section demonstrates how machine learning results can be integrated into interactive dashboards to support business intelligence and strategic decision-making.

6.5 Interactive Data Exploration

One of the key advantages of the Streamlit dashboard is the ability to interactively explore the data.

The dashboard includes several interactive components that allow users to dynamically explore different aspects of the dataset. These features include:

- Navigation menus for switching between different analytical modules
- Interactive charts and visualizations
- Dataset viewing options for examining detailed loan records
- Dynamic filtering and selection options

These interactive capabilities allow users to analyze the data from multiple perspectives without needing to run complex analytical scripts.

The use of Streamlit enables the entire analytical workflow—from data preprocessing to machine learning segmentation—to be presented within a single interactive platform.

Chapter 7

Results and Business Insights using Performance Clusters

After performing the clustering analysis on the branch-level dataset, the branches were segmented into four distinct operational groups based on their performance characteristics. The segmentation was derived using the K-Means clustering algorithm described in Chapter 5. The clustering process relied on three key indicators that represent the core dimensions of gold loan branch operations:

- **Business Volume** – measured using Total Loan Disbursement
- **Profitability** – measured using Average Rate of Interest (ROI)
- **Risk Exposure** – measured using Average Loan-to-Value Ratio (LTV)

These three variables collectively capture the fundamental aspects of gold loan operations. Business volume reflects the scale of lending activity handled by each branch, profitability indicates the financial return generated through lending, and risk exposure represents the level of collateral coverage associated with the loans issued.

Using these indicators, the clustering model identified four performance segments that categorize branches according to their operational behaviour:

1. **High Performance – Low Risk Branches**
2. **High Performance – High Risk Branches**
3. **Stable Branches**
4. **Underperforming Branches**

Each of these segments represents a different operational profile in terms of business growth, lending practices, and financial risk exposure. Understanding these segments is important for identifying strengths within the branch network as well as areas where improvements may be required. The segmentation also allows management to design targeted strategies for improving branch performance and optimizing lending operations.

7.1 Branch Distribution Across Segments

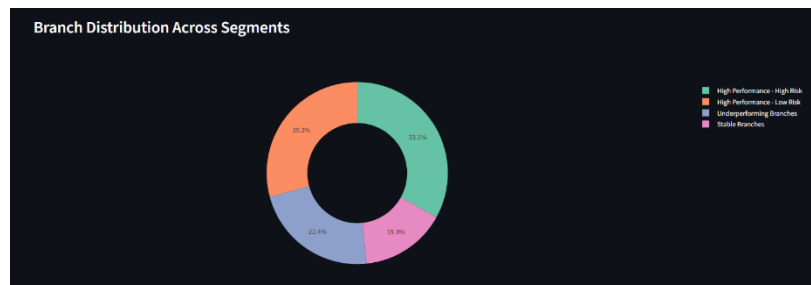


Figure 7.1: Branch distribution across the four performance segments.

This figure illustrates the overall distribution of branches across the four identified performance categories. The visualization provides a clear understanding of how the organization's branch network is distributed in terms of performance and risk exposure.

From the chart, the approximate distribution of branches across the segments is as follows:

- **High Performance – High Risk:** approximately 33.1%
- **High Performance – Low Risk:** approximately 29.2%
- **Underperforming Branches:** approximately 22.4%
- **Stable Branches:** approximately 15.3%

This distribution reveals important information about the structure of the organization's lending operations.

Observations

- The largest portion of branches belongs to the High Performance – High Risk category, which indicates that a significant number of branches generate strong loan disbursement volumes but operate with relatively higher lending risk levels.
- Nearly one-third of the branches operate in environments where loan demand is high, leading to aggressive lending activity and higher LTV ratios.
- The High Performance – Low Risk category forms the second-largest segment, representing branches that successfully maintain strong business activity while controlling risk exposure.
- The Stable Branch segment represents a smaller proportion of the network, indicating moderate operational performance and balanced risk levels.
- Around 22% of the branches fall under the Underperforming category, highlighting the presence of branches that may require operational improvements or strategic support.

Business Implications

- The large share of high-performing branches suggests that the gold loan market is actively utilized in several regions, providing strong business opportunities.
- However, the presence of a large high-risk segment also indicates the need for continuous monitoring of lending practices.
- Identifying underperforming branches allows the organization to focus on targeted interventions and performance improvement strategies.

7.2 High Performance – Low Risk Branches

High Performance – Low Risk branches represent the most efficient and desirable operational category in the branch network. These branches demonstrate strong lending activity while maintaining controlled levels of risk exposure.

Branches in this segment typically show the following characteristics:

- High loan disbursement volumes
- Strong profitability levels
- Well-managed LTV ratios
- Balanced risk-return relationship

These branches are able to maintain high levels of lending activity without significantly increasing financial risk, indicating effective branch management and disciplined lending practices.

Profitability Comparison

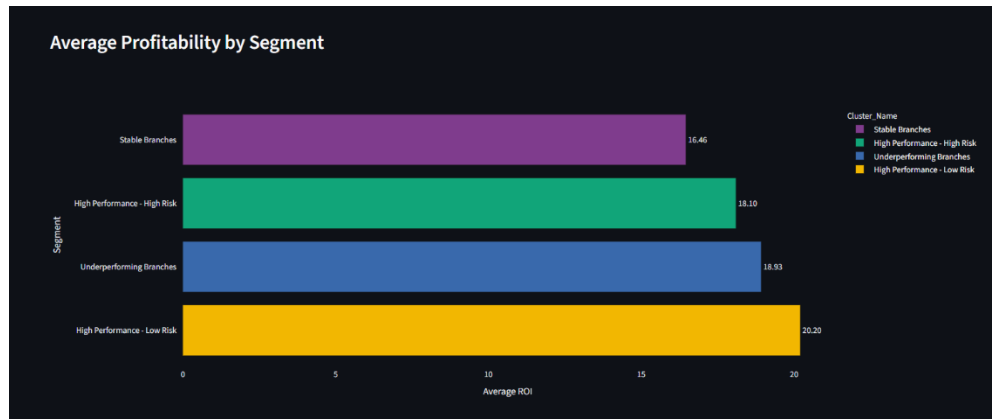


Figure 7.2: Average profitability (ROI) across different branch segments.

This visualization compares the average rate of interest generated by each cluster of branches. ROI represents the profitability of the loans issued by the branches.

From the chart, it can be observed that:

- High Performance – Low Risk branches generate the highest average ROI (approximately 20.2%)
- Underperforming branches generate an ROI of approximately 18.9%
- High Performance – High Risk branches generate around 18.1% ROI
- Stable branches show the lowest ROI (approximately 16.4%)

Interpretation

The higher ROI observed in High Performance – Low Risk branches suggests that these branches operate with efficient loan pricing strategies and strong operational performance.

Possible factors contributing to this performance include:

- Effective customer targeting and loan scheme utilization
- Strong demand for gold loans in the regions where these branches operate
- Efficient loan processing and operational management
- Balanced risk management policies

Strategic Importance

These branches serve as benchmark models for operational excellence within the organization. Their ability to maintain high profitability while controlling risk makes them valuable examples of effective branch management.

Strategic Recommendations

- Analyse operational strategies followed by these branches and replicate them across other branches.
- Use these branches as reference models for training and performance improvement initiatives.
- Strengthen similar lending strategies in other high-potential regions.

7.3 High Performance – High Risk Branches

High Performance – High Risk branches represent branches that generate strong business volumes but operate with relatively higher lending risk.

Branches in this segment typically demonstrate:

- High loan disbursement volumes
- High loan demand in their operational regions
- Higher average LTV ratios
- Strong revenue generation

These branches contribute significantly to the organization's overall business growth, but they also require careful risk management.

Risk vs Performance Relationship

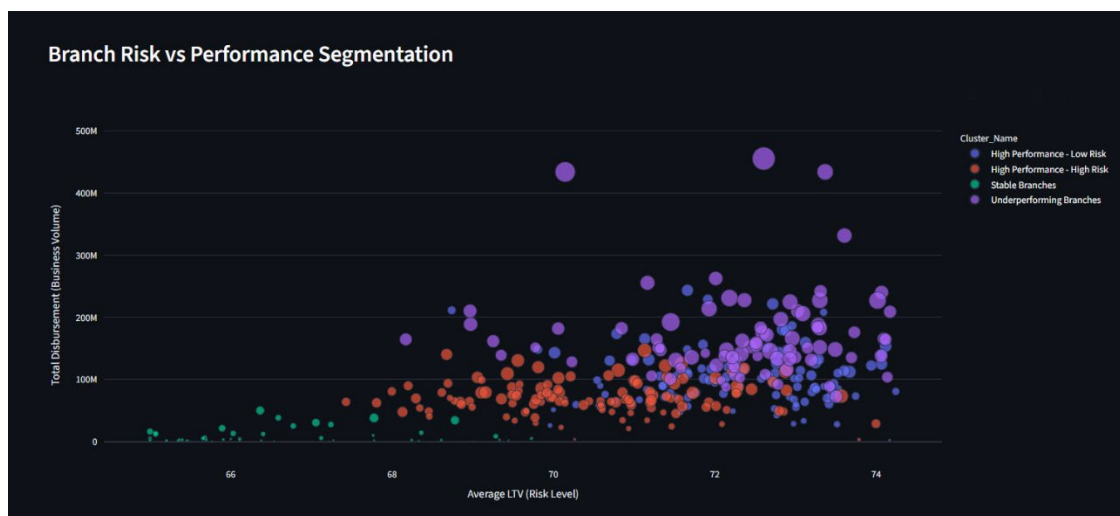


Figure 7.3: Risk versus business volume segmentation of branches.

This scatter plot illustrates the relationship between Average LTV (risk level) and Total Disbursement (business volume) across different clusters.

Observations

- Branches with higher disbursement volumes are often associated with higher LTV ratios, indicating a trade-off between business growth and risk exposure.
- The High Performance – High Risk cluster appears prominently in regions where LTV values range between 72% and 74%.

- Several branches in this cluster demonstrate very high disbursement volumes, suggesting strong loan demand in these regions.

Interpretation

The positive relationship between business volume and risk level indicates that branches with higher loan demand may adopt more aggressive lending policies to attract customers.

While this strategy can increase revenue in the short term, it may also expose the organization to greater financial risk if market conditions change.

Strategic Considerations

- Implement risk monitoring mechanisms for branches in this segment.
- Establish limits for LTV ratios to maintain financial stability.
- Monitor gold price fluctuations that may affect collateral valuation.
- Ensure that lending policies remain aligned with organizational risk management guidelines.

7.4 Stable Branches

Stable branches represent branches with moderate performance levels and balanced operational characteristics. These branches operate consistently but may not generate extremely high loan volumes.

Branches in this category generally show:

- Moderate loan disbursement volumes
- Balanced risk exposure
- Stable but relatively lower profitability levels

Business Volume Comparison

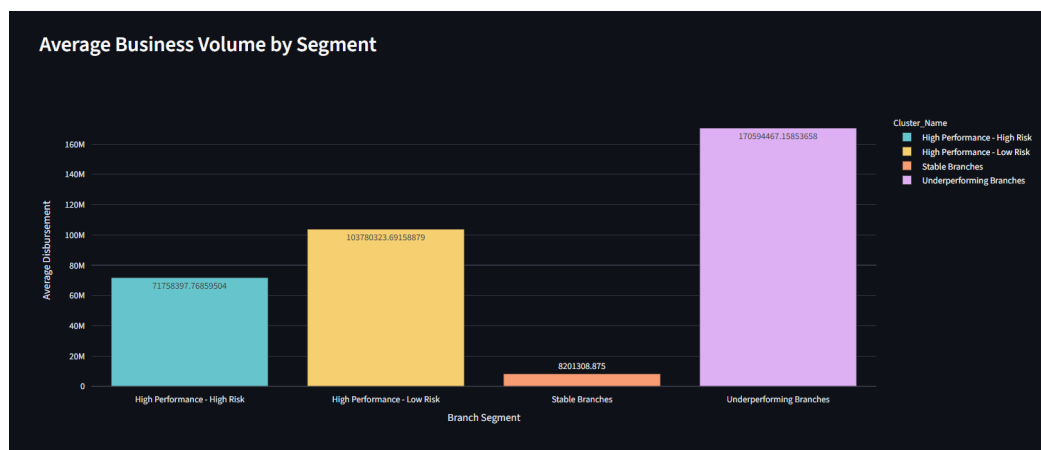


Figure 7.4: Average business volume across branch segments.

This visualization compares the average disbursement volume across the different segments.

From the chart:

- Underperforming branches show the highest average disbursement (~₹170M)
- High Performance – Low Risk branches show approximately (~₹104M)
- High Performance – High Risk branches show (~₹71M)
- Stable branches show the lowest (~₹8M)

Interpretation

Stable branches maintain moderate operational activity and generally operate in regions with steady but limited loan demand.

These branches often serve as supporting operational units within the branch network, maintaining consistent but moderate lending activity.

Risk Comparison

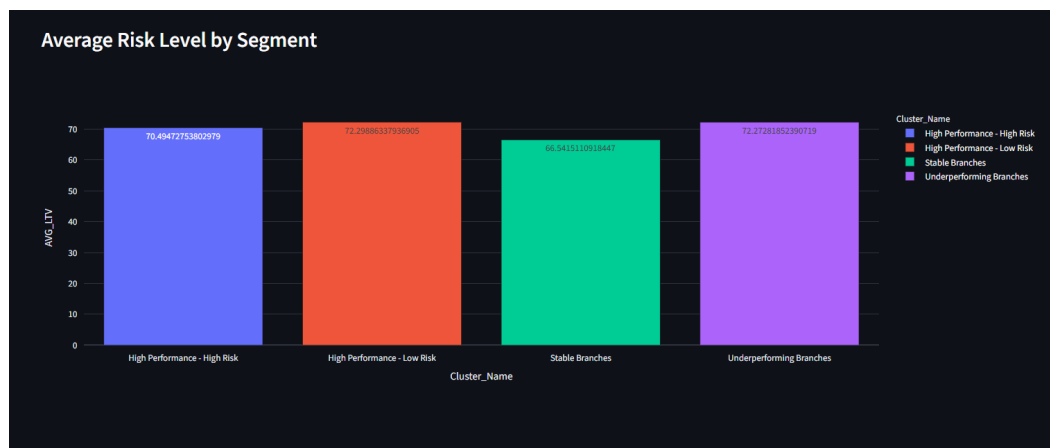


Figure 7.5: Average lending risk (LTV) across branch segments.

This chart compares the average LTV values across segments.

From the chart:

- Stable branches show the lowest average LTV (~66.5%), indicating conservative lending policies.
- Lower LTV ratios reduce financial risk but may also limit loan disbursement potential.

Strategic Opportunities

- Stable branches offer potential for controlled business expansion.
- Marketing efforts and promotional campaigns may help increase customer demand.
- Adjusting loan schemes in these regions may help improve lending activity while maintaining balanced risk.

7.5 Underperforming Branches

Underperforming branches represent branches with relatively lower operational efficiency and business activity.

These branches may exhibit:

- Lower loan disbursement volumes
- Moderate profitability levels
- Higher risk exposure in some cases

Possible Reasons for Underperformance

Several factors may contribute to the lower performance of these branches:

- Limited demand for gold loans in certain regions
- High competition from other financial institutions
- Inefficient branch operations
- Limited awareness of available loan schemes among customers

Observations

Despite lower business volumes, some underperforming branches still maintain moderate ROI levels. This indicates that profitability alone is not sufficient to ensure strong operational performance.

Improvement Strategies

- Strengthen marketing efforts in low-performing regions.
- Introduce region-specific loan schemes to attract more customers.
- Improve branch operational efficiency through staff training and process improvements.
- Enhance customer engagement and service quality.

7.6 Strategic Business Insights

The clustering-based segmentation provides several valuable insights that can support strategic decision-making.

Key Insights

- Maintaining a balance between profitability and risk exposure is essential for sustainable lending operations.
- High Performance – Low Risk branches represent the most efficient operational model within the branch network.
- High Performance – High Risk branches contribute significantly to business growth but require careful risk monitoring.
- Stable branches provide opportunities for gradual expansion of lending activities.

- Underperforming branches require targeted operational improvements and strategic interventions.

Organizational Benefits

By applying machine learning-based segmentation, the organization can:

- Identify high-performing branches
- Monitor risk exposure across the branch network
- Allocate resources more effectively
- Improve overall lending efficiency

The insights obtained from this analysis demonstrate how data-driven decision-making can enhance operational performance in financial institutions.

Chapter 8

Conclusion and Future Scope

This chapter summarizes the overall findings of the project and evaluates the outcomes achieved through the analysis of the gold loan disbursement dataset. The project focused on analyzing loan transaction data, evaluating branch-level performance, and applying machine learning techniques to segment branches based on operational characteristics.

Through the integration of data preprocessing, exploratory data analysis, branch-level aggregation, machine learning clustering, and interactive dashboard visualization, the project successfully transformed raw transactional data into meaningful business insights. The results demonstrate how data analytics and machine learning techniques can support operational decision-making in financial institutions.

8.1 Conclusion

The main objective of this project was to analyze gold loan disbursement data in order to evaluate branch performance and identify operational patterns across the branch network. By applying data analytics and machine learning techniques, the study was able to uncover valuable insights related to loan disbursement behavior, profitability levels, and lending risk exposure.

The analysis began with data preprocessing and cleaning of the gold loan transaction dataset, followed by exploratory data analysis to understand loan distribution across different states, schemes, and branches. These analyses provided an initial understanding of how gold loan operations vary across different geographical regions and product schemes.

Subsequently, branch-level aggregation was performed to calculate key performance indicators such as total disbursement, loan count, average ROI, average LTV, and average gold weight. These metrics allowed the evaluation of operational performance at the branch level rather than the individual loan level.

A machine learning-based clustering model was then applied to segment branches based on three key factors: business volume (total disbursement), profitability (ROI), and lending risk (LTV). The clustering analysis successfully categorized branches into four meaningful performance groups:

- High Performance – Low Risk Branches
- High Performance – High Risk Branches
- Stable Branches
- Underperforming Branches

These segments provide a structured understanding of branch behavior and highlight differences in operational performance across the branch network.

In addition to the analytical models, an interactive Streamlit-based dashboard was developed to visualize the analysis results. The dashboard integrates loan-level analytics, branch performance analysis, and clustering insights into a user-friendly platform that enables interactive data exploration.

Overall, the project demonstrates how data analytics and machine learning can be applied to financial operational data to support better performance monitoring and strategic decision-making.

8.2 Objectives Achieved

The objectives defined at the beginning of the project were successfully achieved through the implementation of various analytical techniques and visualization tools.

The first objective was to analyze loan disbursement patterns across branches, states, and schemes. This was accomplished through exploratory data analysis and visualizations that highlighted geographical distribution, scheme usage patterns, and loan characteristics.

The second objective was to evaluate branch-level operational performance using key performance indicators such as ROI, LTV, and total disbursement. Through branch-level aggregation and performance analysis, it was possible to identify differences in lending behaviour across branches.

The third objective was to identify high-performing and underperforming branches using analytical methods. This objective was achieved through clustering analysis, which grouped branches into meaningful segments based on performance and risk characteristics.

Another objective of the project was to measure lending risk using the loan-to-value (LTV) ratio. By analyzing LTV distributions across branches and clusters, the study was able to identify areas where lending risk is relatively higher.

Finally, the project aimed to build a visual analytics platform for presenting the analysis results. This objective was achieved through the development of a Streamlit dashboard, which allows users to interactively explore loan data, branch performance metrics, and clustering results.

8.3 Limitations of the Study

Although the project successfully demonstrates the application of data analytics and machine learning techniques to gold loan operations, several limitations should be considered when interpreting the results.

One of the primary limitations of the study is the limited time span of the dataset. The dataset used in this project covers only approximately 45 days of loan transaction data. Because of this short time period, the dataset may not fully capture long-term operational trends, seasonal patterns, or fluctuations in loan demand.

Gold loan operations can vary over time due to factors such as:

- Seasonal financial needs of customers
- Changes in gold market prices
- Regional economic conditions
- Promotional loan schemes offered by financial institutions

Since the dataset represents only a short observation window, the results of the analysis may not reflect the complete operational behavior of the branch network over longer time periods.

Another limitation relates to the scope of available variables in the dataset. While the dataset includes key financial parameters such as loan amount, LTV ratio, and ROI, other potentially useful variables such as customer demographics, repayment behavior, and loan renewal patterns were not included. The absence of these variables limits the ability to perform deeper behavioral analysis.

Additionally, clustering analysis depends heavily on the selected features. Although ROI, LTV, and total disbursement provide meaningful indicators of performance, incorporating additional features could potentially improve the segmentation results.

Despite these limitations, the study still provides valuable insights into branch performance and demonstrates the potential of data analytics techniques in financial operations analysis.

8.4 Future Enhancements

There are several opportunities to extend and improve this study in future work.

One possible enhancement is the use of larger and longer-term datasets. By incorporating data covering multiple months or years, it would be possible to analyze long-term trends, seasonal lending patterns, and changes in branch performance over time.

Future studies could also incorporate additional variables such as customer demographics, loan repayment behavior, and loan renewal patterns. Including these variables would allow more comprehensive analysis of customer behavior and lending risk.

Another potential improvement is the application of advanced machine learning models. While this study used clustering techniques for segmentation, future research could explore predictive models for forecasting loan demand, predicting branch performance, or identifying potential risk exposure.

The dashboard developed in this project could also be expanded further. Additional features such as dynamic filtering, predictive analytics, and automated reporting tools could enhance the usability of the dashboard for business users.

Furthermore, integrating the dashboard with real-time data sources could enable continuous monitoring of branch performance and allow management to make faster and more informed decisions.

8.5 Summary

This project demonstrates how data analytics and machine learning techniques can be applied to financial transaction data to generate meaningful business insights. By analyzing gold loan disbursement data, evaluating branch-level performance, and applying clustering-based segmentation, the study provides a structured understanding of branch operations.

The results of the analysis highlight differences in business volume, profitability, and risk exposure across branches, allowing the identification of high-performing, stable, and underperforming branches.

The development of an interactive Streamlit dashboard further enhances the practical value of the project by providing an accessible platform for exploring analytical results.

Although the dataset has certain limitations, the overall findings demonstrate the potential of data-driven approaches for improving operational performance and strategic decision-making in financial institutions.

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